TECHNICAL BRIEF: OptiqAI PhotonPilot

Accelerating the Photonic Tapeout Cycle with AI-Native Design

The Problem: The "Simulation Tax"

In 2026, Silicon Photonics is the bottleneck for AI scaling. Current PDA (Photonic Design Automation) tools rely on 30-year-old FDTD solvers. A single optimization loop for a high-efficiency grating coupler takes **45–60 minutes**, leading to weeks-long design cycles for a single PIC.

The Solution: Neural Surrogate Engines

OptiqAI replaces traditional iterative solvers with a **Physics-Informed Neural Operator (PINO)** architecture. Our platform allows engineers to draw a component and see the electromagnetic field profile and S-parameters **instantly**.

Benchmarking: OptiqAI vs. Traditional FDTD (Meep)

Component	Metric	Meep (3D-FDTD)	OptiqAI (Surrogate)	Speedup
Ring Resonator	Compute Time	38 Minutes	**~120 ms**	**19,000x**
	Accuracy (R^2)	1.0 (Ref)	**0.988**	-
Grating Coupler	Compute Time	52 Minutes	**~180 ms**	**17,300x**
	Accuracy (R^2)	1.0 (Ref)	**0.982**	-

Testing conducted on 220nm SOI platform at 1550nm wavelength. AI results obtained via single GPU inference.

Core Innovation & Moat

1. **Agent-Native Infrastructure:** Built-in **AI Copilot** that converts natural language intent (e.g., *"Optimize this MZI for 1310nm with <1dB loss"*) into optimized GDSII geometry using Adjoint-based inverse design.

2. **Collaborative Cloud CAD:** Real-time multi-user synchronization (Y.js/CRDT) allows global teams to co-design layouts in a single browser tab.
3. **Security & Deployment:** Native support for **Object Storage (S3/MinIO)** and **Private Cloud (VPC)** deployment, ensuring proprietary IP never leaves the customer's secure perimeter.

Founding Team & Traction

- **Status:** Maturing MVP; Functional WebGL Layout Engine + AI field predictor.
- **Network:** Currently participating in the **a16z Talent Network**.
- **Traction:** 20+ verified waitlist signups from Tier 2 Photonics startups and academic labs.